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About the UltraWave24 MWReceiver : Features

Features

The MWReceiver instrument accurately captures signals with frequency components from 50MHz to 6GHz. The downconverted signal is captured by a dedicated digitizer on a MW Measure board. Basic functional specifications of the UltraWave24 MWReceiver are:

- Frequency range: 50MHz to 6GHz
- Input amplitude range: -130dBm to +30dBm
- Two digitizers:
  - VHF (5MHz-165 MHz bandwidth, 12 bit)
  - High speed (500 kHz-45MHz bandwidth, 16 bit)
- Programmable RF attenuation and IF gain boost for maximum dynamic range
- Selectable IF filters
- Selectable on-board signal processing with a hardware DFT
- Optional conversion of signals to complex (IQ) domain for digital demodulation

For UltraWave24 hardware specifications, see UltraWave24 Specifications.

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About the UltraWave24 MWReceiver : Safety Information

Safety Information

When operating the Microwave Receiver instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraWave24 MWReceiver : Key Terms

Key Terms

IF Output	Signal path from the IF signal right after the receiver downconverter to a DIB port.
IF	Intermediate frequency
LO	Local Oscillator
DFT	Discrete Fourier Transform. The MWReceiver has a DFT implemented in hardware.
RF	Radio frequency
QDC	Quadrature Down Converter. The HS ADC of the MWReceiver can shift an IF spectrum to DC in hardware, allowing smaller captures and faster results processing.

For more MW terms and definitions, see Key Terms in About the UltraWave24 Subsystem.

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About the UltraWave24 MWReceiver

About the UltraWave24 MWReceiver

The MWReceiver captures signals with frequency components from 50MHz to 6GHz. Captured data is processed and analyzed on the DSP subsystem or the on-board DFT hardware.The MWReceiver can also send the downconverted signal to a DIB port.

The Microwave Receiver instrument is designed to capture high frequency signals that are used in Microwave/RF telecommunications applications. Typical tests that can be performed are RF power measurements, gain, and IP3.

This section includes the following topics:

- Features
- Safety Information
- Key Terms

Microwave Receiver information includes the following sections:

- UltraWave24 MWReceiver Theory of Operation
- UltraWave24 MWReceiver Programming
- UltraWave24 MWReceiver Debug Display

Related Topics

- UltraWave24 Specifications
- MWReceiver (VBT syntax)
- UltraWave24 DIB Design Guidelines
- About the UltraWave24 Subsystem
- UltraWave24 Examples
- Vector Scope

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UltraWave24 MWReceiver Debug Display

UltraWave24 MWReceiver Debug Display

The Microwave Debug Display can display one or more instrument blocks for the MWReceiver logical instrument. Each instrument block shows the state of an instrument in the context of a site.

This section includes the following topic:

- UltraWave24 MWReceiver Debug Display Controls

For an overview of all displays supporting the UltraWave24 subsystem, see UltraWave24 Displays.

For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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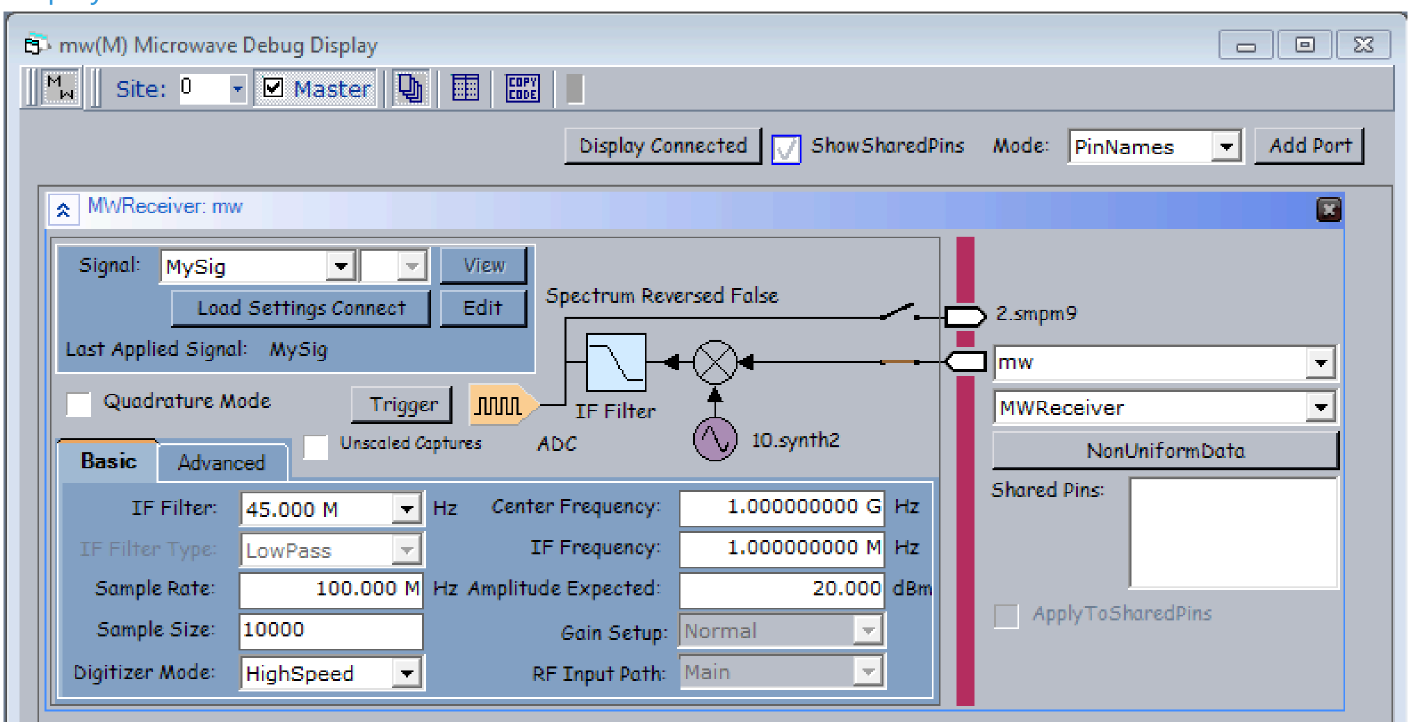
UltraWave24 MWReceiver Debug Display : UltraWave24 MWReceiver Debug Display Controls

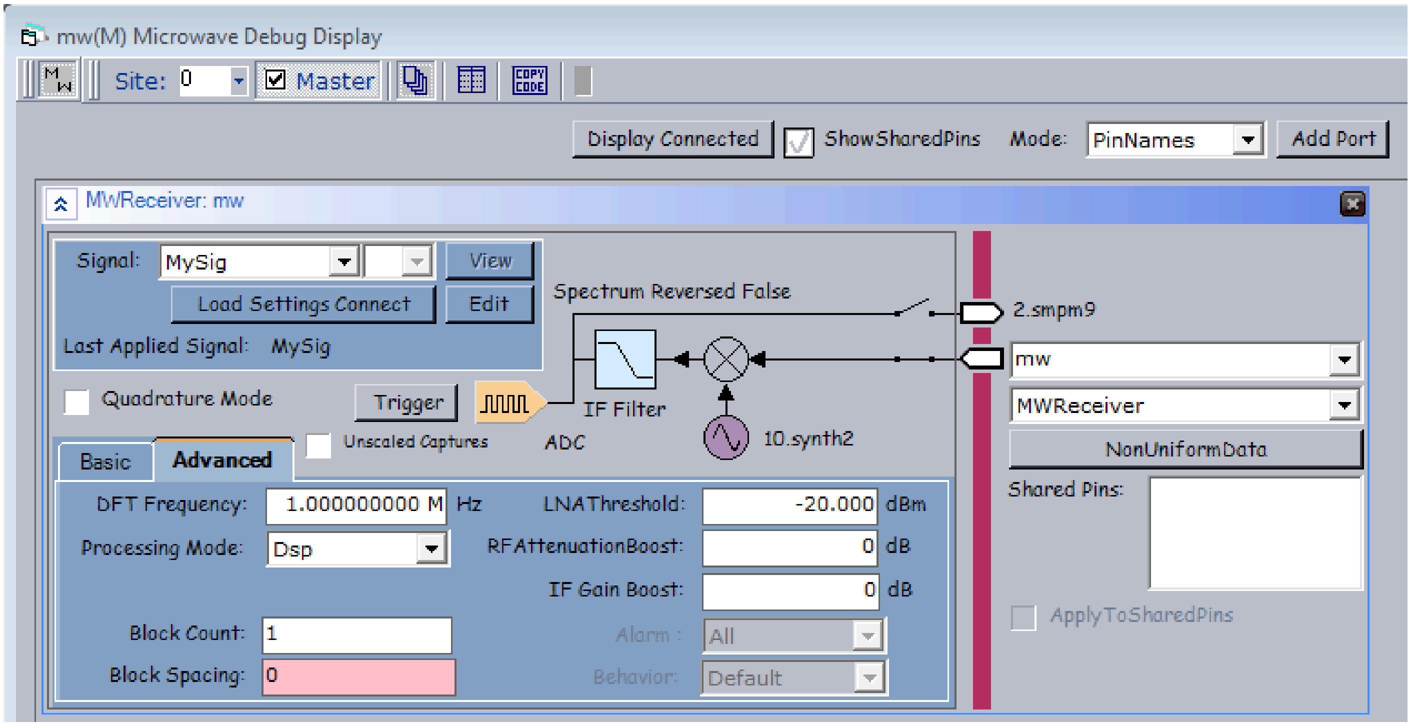
UltraWave24 MWReceiver Debug Display Controls

MWReceiver port display controls allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWReceiver VBT language.

For information about the common debug display controls, see UltraWave24 Microwave Debug Display.



UltraWave24 MWReceiver Logical Instrument Display, Basic Tab



UltraWave24 MWReceiver Logical Instrument Display, Advanced Tab

The following controls are available on the MWReceiver logical instrument block:

Signal	Name of the signal used by the source. Available signals appear on the drop-down list. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
Signal index	Displays the selected Signal index on a drop-down list of available indexes.
View	Displays a list of signal viewer tools (such as WaveScope or VectorScope) for the selected signal on a drop-down list of available viewers.
Load Settings Connect	Loads the selected signal’s timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument’s settings and connects the MWReceiver.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Info	Displays information for the selected signal in a pop-up balloon window.
Last Applied Signal	Name of the last signal applied to the source.

Quadrature Mode	Enables the quadrature mode, when checked.
Trigger	Triggers a capture.
Unscaled Capture	Enables or disables autoscaling. Unscaling applies to all captures.
Spectrum Reversed	Displays whether the spectrum is reversed in the capture.
synth label	Identifies the synthesizer (slot and number) used by this MWReceiver logical instrument.
connection relay	The relay control connects or disconnects the MWReceiver logical instrument to the specified pin or channel. Click the relay to open or close the connection. In addition to the input terminal, the MWReceiver has an IF output terminal to route the IF output to the DIB.
Basic tab	
IF Filter	Displays or sets the IF filter frequency on a drop-down list.
IFFilter Type	Displays and sets the IF filter type on a pull-down list. Grayed out for UltraWave24 MWReceiver.
Sample Rate	Displays and sets the digitizer sample rate. The tooltip displays the allowable sample rate range.
Sample Size	Displays and sets the capture sample size. The sample size is for a single block. The tooltip displays the allowable sample size range.
Digitizer Mode	Displays and sets the digitizer mode on a drop-down list: • HighSpeed • VeryHighSpeed You cannot change the digitizer mode unless all digitizer parameter values are within the range of the very high speed digitizer.
Center Frequency	Displays and sets the MWReceiver input center frequency. The tooltip displays the allowable center frequency range.
IF Frequency	Displays and sets the downconverter IF frequency. The MWReceiver uses the IF frequency and the center frequency to set the LO frequency for the downconverter mixer.

Amplitude Expected	Displays and sets the expected signal amplitude (power) at the port. The expected amplitude is for the full 50MHz to 6GHz bandwidth. The tooltip displays the allowable amplitude (power) range.
Gain Setup	Displays and sets the gain setup for a modulated tone. Grayed out for UltraWave24 MWReceiver.
RF Input Path	Displays and sets the MWReceiver input path to the main path (through the step attenuator) or the alternate (through the coupler). Grayed out for UltraWave24 MWReceiver.
Advanced tab	
DFT Frequency	Displays and sets the frequency for the single-point DFT performed in the receiver hardware. Used when Processing Mode is set to DFT. The tooltip displays the allowable DFT frequency range.
Processing Mode	Displays and sets the processing mode on a drop-down list: • Dsp: Capture results processed on the DSP subsystem. • DFT: Capture results processed on the hardware single-point DFT processor.
Block Count	Displays and sets the number of blocks of capture data. The tooltip displays the allowable block count range.
Block Spacing	Displays and sets the number of empty samples between blocks of capture data. The tooltip displays the allowable block spacing range.
LNA Threshold	Displays and sets the LNA threshold.
RF Attenuation Boost	Displays and sets the RF attenuation increase.
IF Gain Boost	Displays and sets the gain increase in the IF section after the filters in the receiver. When you set the value, you request a value of gain boost. The displayed value is the actual amount of gain boost provided by the hardware. The tooltip displays the allowable IF gain boost range.
Alarm	Displays and sets the MWReceiver alarm type on a drop-down list.
Behavior	Displays and sets the MWReceiver alarm behavior on a drop-down list.

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UltraWave24 MWReceiver Programming

UltraWave24 MWReceiver Programming

The MWReceiver is a MW logical instrument in the IG-XL environment. You add the MWReceiver to an IG-XL test program in the same general way that you add other MW instruments.

The following topics are available:

- Capturing RF Signals Using the MWReceiver
- MWReceiver Pattern Control

Before writing test code using the MWReceiver, see UltraWave24 Instrument Programming.

For MWReceiver concepts, a programming procedure, and an example, see Capturing RF Signals Using the MWReceiver.

For more information, see:

- MWReceiver (VBT Syntax Reference)
- UltraWave24 Examples

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UltraWave24 MWReceiver Programming : Capturing RF Signals Using the MWReceiver

Capturing RF Signals Using the MWReceiver

This section includes the following topics:

- Programming Concepts
- Example

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UltraWave24 MWReceiver Programming : Capturing RF Signals Using the MWReceiver : Programming Concepts

Programming Concepts

This section contains the following MWReceiver programming concepts:

MWReceiver Reset Values

The table lists the local reset values for the MWReceiver.

MWReceiver Reset Values

Setting	Reset Value
Amplitude Expected	20dBm
Center Frequency	1.0GHz
IF Frequency	1.0MHz

IF Filter	45.0MHz
Sample Rate	100.0MHz
Sample Size	10000
Digitizer Mode	HighSpeed
Quadrature Mode	Off
DFT Frequency	1.0MHz
Processing Mode	DSP
IF Gain Boost	0dB
Block Count	1
Block Spacing	0
Connections	Disconnected
RF Attenuation Boost	0dB
LNA Threshold	-20dBm

MWReceiver Gain Calibration

Receive run-time calibration (leveling) is affected by any change to the following receiver parameters:

- AmplitudeExpected
- CenterFrequency
- DigitizerMode
- IFFilter
- IFFrequency
- IFGainBoost
- RFAttenuationBoost

IF Spectral Inversion

Due to the operation of the MWReceiver, the IF spectrum created from captured samples can be inverted if the receiver sets a “high-side” LO. A high-side LO is an LO frequency higher than the RF input to the mixer.

When the IF spectrum is inverted (reversed), frequency bins are arranged from high to low, and you must account for it in bin calculations for the frequencies of interest.

You can use the Pins.IsSpectrumReversed property to determine whether the MWReceiver LO setting for a specific frequency will result in an inverted spectrum.

Quadrature Downconversion

A quadrature downconverter (QDC) in the MWReceiver converts a digitized IF signal from the HS ADC to a complex signal at DC (0Hz). (The QDC is not used with the VHF ADC.) A QDC allows significantly smaller captures and the resulting faster DSP processing, while retaining the same frequency resolution.

You can use the .QuadratureMode property to turn on the MWReceiver quadrature downconverter. For more information, see TheHdw.MWReceiver.Pins.QuadratureMode.

TSD File Support

The MWReceiver accepts signal definitions for modulated signals in TSD files. TSD files are loaded using the .SignalDefinitionFile property.

For more information, see:

- Using the ESA Toolkit
- TheHdw.MWReceiver.Pins.Signals.Item.SignalDefinitionFile

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UltraWave24 MWReceiver Programming : Capturing RF Signals Using the MWReceiver : Example

Example

For an example of basic MWReceiver code, see MWSource-MWReceiver Loopback Test.

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UltraWave24 MWReceiver Programming : MWReceiver Pattern Control

MWReceiver Pattern Control

You can control an MWReceiver instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- MWReceiver Pattern Microcodes
- MWReceiver ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control the MWReceiver on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave24 MWReceiver Programming : MWReceiver Pattern Control : MWReceiver Pattern Microcodes

MWReceiver Pattern Microcodes

A microcode controls a single function, such as starting the capture of a signal. A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is captured. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start capturing a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used for issuing commands to an MWReceiver instrument from a pattern.

MWReceiver Pattern Microcodes

Microcode	Functional Description
Disconnect	Instructs an MWReceiver to reset all connections to their disconnected state.
Trig <SignalName>	The instrument immediately starts capturing the named capture signal. The Trig microcode takes an optional argument, which is the name of the signal to be captured. If the signal name is unspecified, the default signal is used. See TheHdw.MWReceiver.Pins.Signals.DefaultSignal.
Resync	Forces phase alignment of an MWReceiver instrument clock with the HSD T0 clock. See Use of Resync.

Enable_Inst_Cond	Enables the instrument condition. The MWReceiver asserts the instrument condition to the pattern generator at the completion of a capture.
Disable_Inst_Cond	Disables an MWReceiver from asserting the instrument condition.
Enable_Alarm	Enables all MWReceiver alarms and opens the alarm window for MWReceiver in a pattern. This microcode reopens the alarm window that had been previously issued a Disable_Alarm microcode.
Disable_Alarm	Disables all MWReceiver alarms and closes the alarm window for MWReceiver in a pattern.
LoadSettingsConnect <SignalName>	Instructs an MWReceiver to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.

Notes on MWReceiver Microcodes

Note the following when using microcodes with an MWReceiver:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:
  - Resync: <1s
  - LoadSettingsConnect: 400s (typical)
  - Disconnect: <200s (typical)
  - Trig: <10s

Note that this is the time to start the trigger. Acquisition time (SampleSize/SampleRate) is additional.

- An MWReceiver does not use PSets. It provides the equivalent capability using Signals programming in VBT.

- When using an UltraWave logical instrument (such as an MWReceiver), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.
  - IG-XL returns an error if you do not follow this guideline.
  - Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see Using UltraWave24 Instruments with POP.

Use of Resync

The Resync microcode in the Microwave subsystem performs additional tasks when compared to the Resync used with the BBAC and VHFAC instruments. The primary function of Resync for BBAC and VHFAC is to align the analog clock for the analog instrument with the digital clock supplying the timing reference for the pattern.

For the Microwave instruments, the Resync microcode also:

- Forces alignment of the internal AWG clocks which drive I and Q modulation
- Synchronizes the start of waveforms

The instantiation time associated with Resync is 1 s, which you must manage. Therefore, you must place a wait of at least 1 s between the Resync and following microcode.

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UltraWave24 MWReceiver Programming : MWReceiver Pattern Control : MWReceiver ASCII Pattern Syntax

MWReceiver ASCII Pattern Syntax

MWReceiver pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs an MWReceiver to perform a particular action, such as triggering a new capture.

You can only use MWReceiver microcodes with pins that the instruments statement has specified as MWReceiver. For example:

```
instruments = {
    DUT_AOUT:MWReceiver;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any MWReceiver related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWReceiver microcode on a vector is:

(pin-list : MWReceiver = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWReceiver. The figure shows this pattern in PatternTool.

```
instruments = {
    Measure:MWReceiver 1;
}

vm_vector
vm_MWReceiveOnly_A ( $tset      , Dig2)
{
start_label MWReceiveOnly:
```

```
                > BasicMeasure  0    ;
(Measure:MWReceiver = Resync)
                > BasicMeasure  0    ;
repeat 10           > BasicMeasure  0    ; //1uS wait time
(Measure:MWReceiver = LoadSettingsConnect GSM)
                > BasicMeasure  0    ;
repeat 10000        > BasicMeasure  0    ; //1mS wait time
(Measure:MWReceiver = Trig GSM)
                > BasicMeasure  0    ;
repeat 10000        > BasicMeasure  0    ; //1mS wait time
(Measure:MWReceiver = LoadSettingsConnect WCDMA)
                > BasicMeasure  0    ;
repeat 10000        > BasicMeasure  0    ; //1mS wait time
(Measure:MWReceiver = Trig WCDMA)
                > BasicMeasure  0    ;
repeat 10000        > BasicMeasure  0    ; //1mS wait time
(Measure:MWReceiver = Disconnect)
                > BasicMeasure  0    ;
repeat 2000        > BasicMeasure  0    ; //200uS wait time
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
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                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
                > BasicMeasure  0    ;
```

```
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
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        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
        > BasicMeasure 0 ;
halt:
        > BasicMeasure 0 ;
}
```

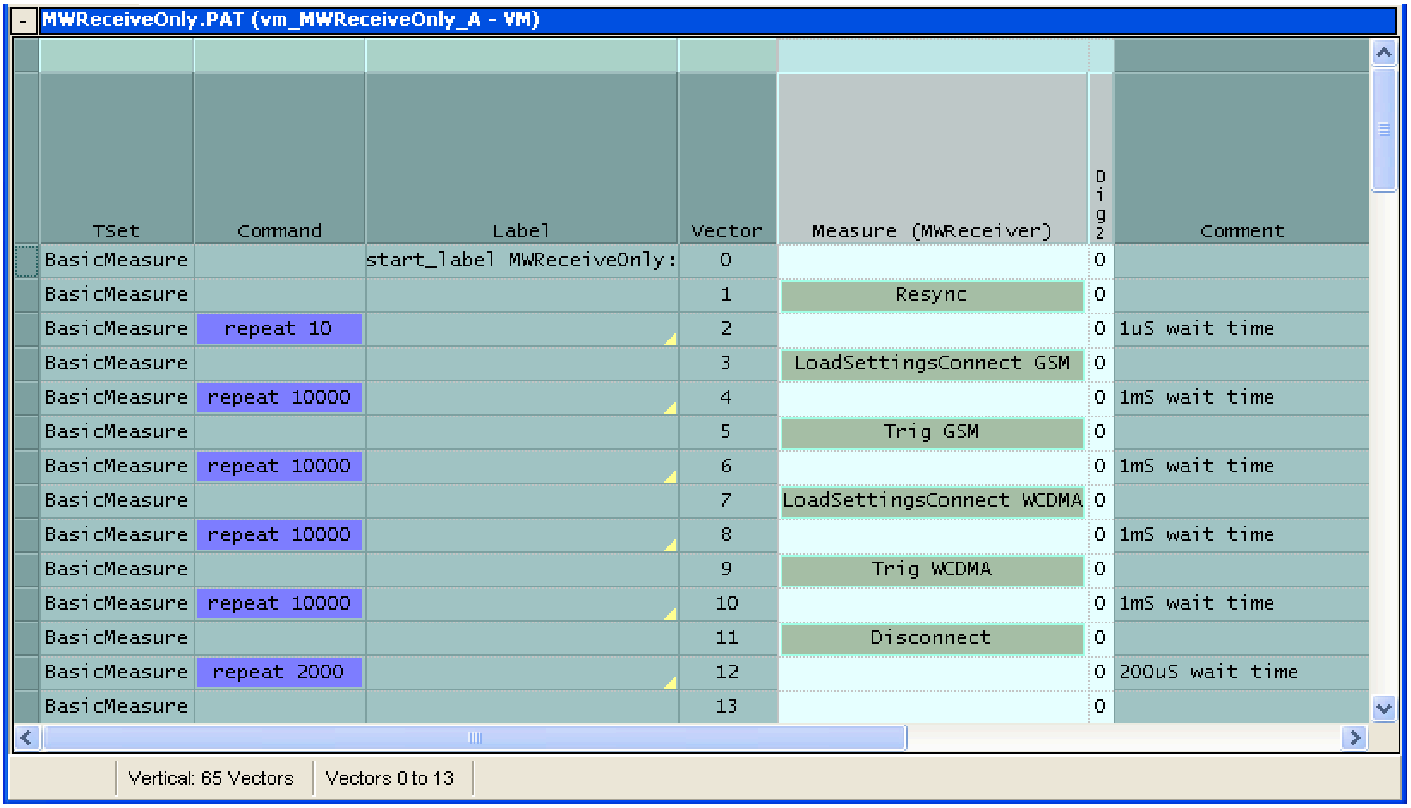

MWReceiveOnly.PAT (vm_MWReceiveOnly_A - VM)

TSet	Command	Label	Vector	Measure (MWReceiver)	Dig2	Comment
BasicMeasure		start_label MWReceiveOnly:	0		0	
BasicMeasure			1	Resync	0	
BasicMeasure	repeat 10		2		0	1uS wait time
BasicMeasure			3	LoadSettingsConnect GSM	0	
BasicMeasure	repeat 10000		4		0	1mS wait time
BasicMeasure			5	Trig GSM	0	
BasicMeasure	repeat 10000		6		0	1mS wait time
BasicMeasure			7	LoadSettingsConnect WCDMA	0	
BasicMeasure	repeat 10000		8		0	1mS wait time
BasicMeasure			9	Trig WCDMA	0	
BasicMeasure	repeat 10000		10		0	1mS wait time
BasicMeasure			11	Disconnect	0	
BasicMeasure	repeat 2000		12		0	200uS wait time
BasicMeasure			13		0	

Vertical: 65 Vectors | Vectors 0 to 13

MWReceiver Pattern Microcodes Example

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UltraWave24 MWReceiver Theory of Operation

UltraWave24 MWReceiver Theory of Operation

The MWReceiver instrument captures RF signals by adjusting the amplitude an input signal at a MW port, downconverting the signal to IF, and then digitizing the IF signal. The MWReceiver has high speed connections to the DSP subsystem for parallel, background DSP processing. In addition, the MWReceiver contains hardware DSP capabilities: a quadrature down converter and fast DFT for single frequency amplitude measurements.

The following topics are available:

- Basic Operation
- MWReceiver Hardware Functions and Signal Flow

For information on MWReceiver instrument performance specifications and standards, see UltraWave24 Specifications.

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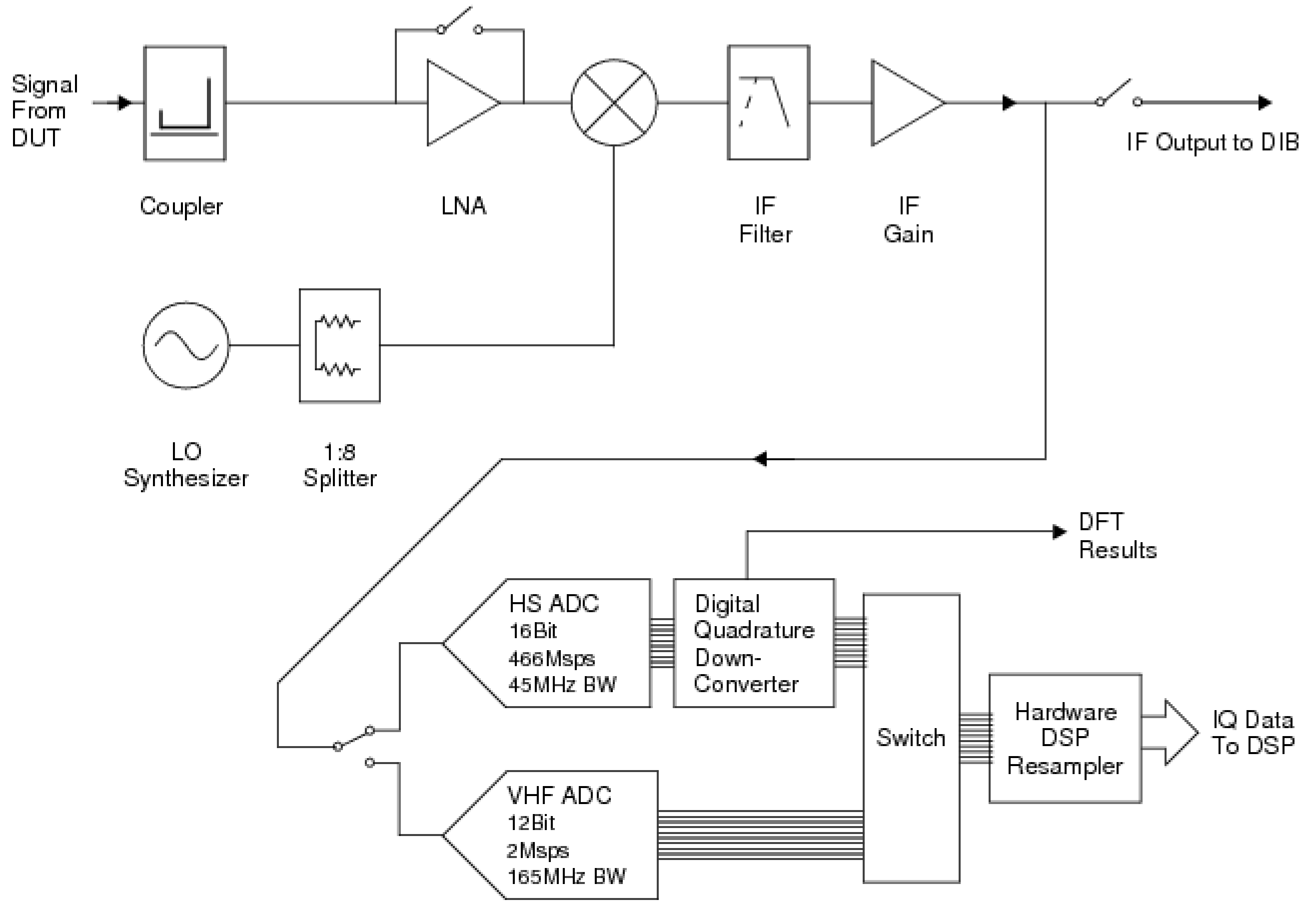
UltraWave24 MWReceiver Theory of Operation : Basic Operation

Basic Operation

At a high level, the MWReceiver hardware can be represented by four stages:

- Input level control (attenuator and LNA)
- Downconverter
- IF filter and gain
- Digitizer, quadrature downconverter, and resampler



UltraWave24 MWReceiver High-Level Block Diagram

As the figure shows, the downconverter accepts an RF input signal from the DUT or a DIB circuit. Before the downconverter, the RF signal level is adjusted as needed using the step attenuator and LNA on the front end.

The downconverter LO is split from the Synthesizer LO by a 1:4 splitter. Since the LO is shared with all of the receiver/downconverters using an LO from a single Synthesizer LO, all MWReceivers must use the same IF frequency.

The downconverted IF signal, in the range from 500kHz to ~600MHz, is routed to selectable IF filters (10MHz low-pass, 22MHz low-pass, or 160MHz low-pass) or a 45MHz low-pass filter by default. The IF gain stage processes the IF signal in 3dB increments and can provide IF gain boost or attenuation.

☑ **Note:**

The 10MHz low-pass filter must be set for the MWReceiver to meet the spur specification for RF frequencies below 100MHz.

If the IF output to DIB relay is closed, the unfiltered IF signal is routed to the DIB.

The IF signal is connected to one of two analog-to-digital converters (ADCs) sampling at a fixed rate. A hardware DSP resamples the data to a programmed sample rate.

The MWReceiver has two digitizers with 8Msamples of capture memory:

- VHF digitizer (165MHz, 12 bits, 116.66667MHz - 466.66667MHz BW)

- High Frequency (HF) digitizer (45MHz, 16 bits, 1.05MHz-133.333333MHz BW)

After conversion and hardware equalization (FIR filter, not shown), the data is sent to a digital quadrature downconverter (QDC). Part of the QDC hardware can perform a DFT and make discrete (single frequency) amplitude measurements very quickly (~300s). This hardware DFT does not require the tester DSP PCs.

Data samples output from the resampler are sent to the DSP subsystem using the high speed Move Link.

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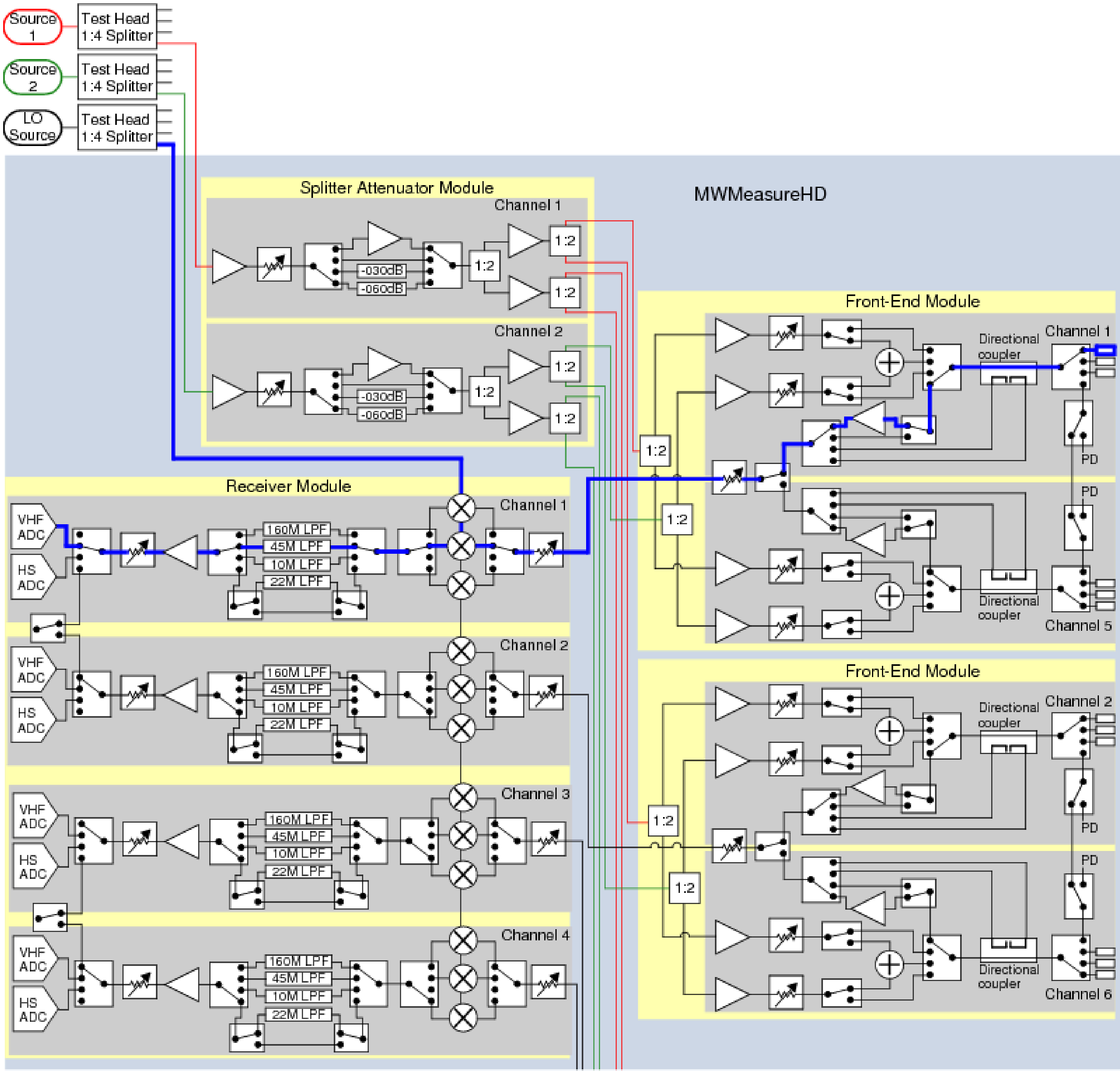
UltraWave24 MWReceiver Theory of Operation : MWReceiver Hardware Functions and Signal Flow

MWReceiver Hardware Functions and Signal Flow

The figure shows the configuration of the MW subsystem and received signal flow for the operation and connection of an MWReceiver. Heavy lines represent the signal flow. See UltraWave24 Function Descriptions for more information about the functions in the diagram.



MWReceiver Hardware Functions and Signal Path, Two of Four FEMs Shown

For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see User View of UltraWave24 SMPM Connector Blocks, User View of UltraWave24 DIB Pad Patterns, and UltraWave24 Signal Descriptions.

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